

Nonparametric regression – curve fitting

The relationship between the effect and the cause is not necessarily linear.

Nonlinear regression requires numerical optimization; therefore, it is more difficult and computationally more intensive to fit than linear regression.

$(X_i, Y_i)_{i=1, \dots, n}$ observation *i.i.d.*

$(x_i, y_i)_{i=1, \dots, n}$ the statistical data.

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Interpolation and smoothing techniques are used to estimate values between our data points $(x_i, y_i)_{i=1, \dots, n}$ and then, to smooth the data.

The conditional expectation is a function with *smoothing* properties:

$$E(X|Y=y) = g(y), \text{ with } g : [a, b] \rightarrow \mathbb{R}$$

$$\exists g'(y), g''(y) \text{ continuous functions}$$

The nonparametric regression curve is:

$$X = g(y) + \xi.$$

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The sum of square deviations with penalty is:

$$SS(g) = \sum_{i=1}^n (x_i - g(y_i))^2 + \alpha \int_a^b (g''(y))^2 dy$$

$\alpha > 0$ is the
smoothing
parameter

it measures the
curvatures in g at y

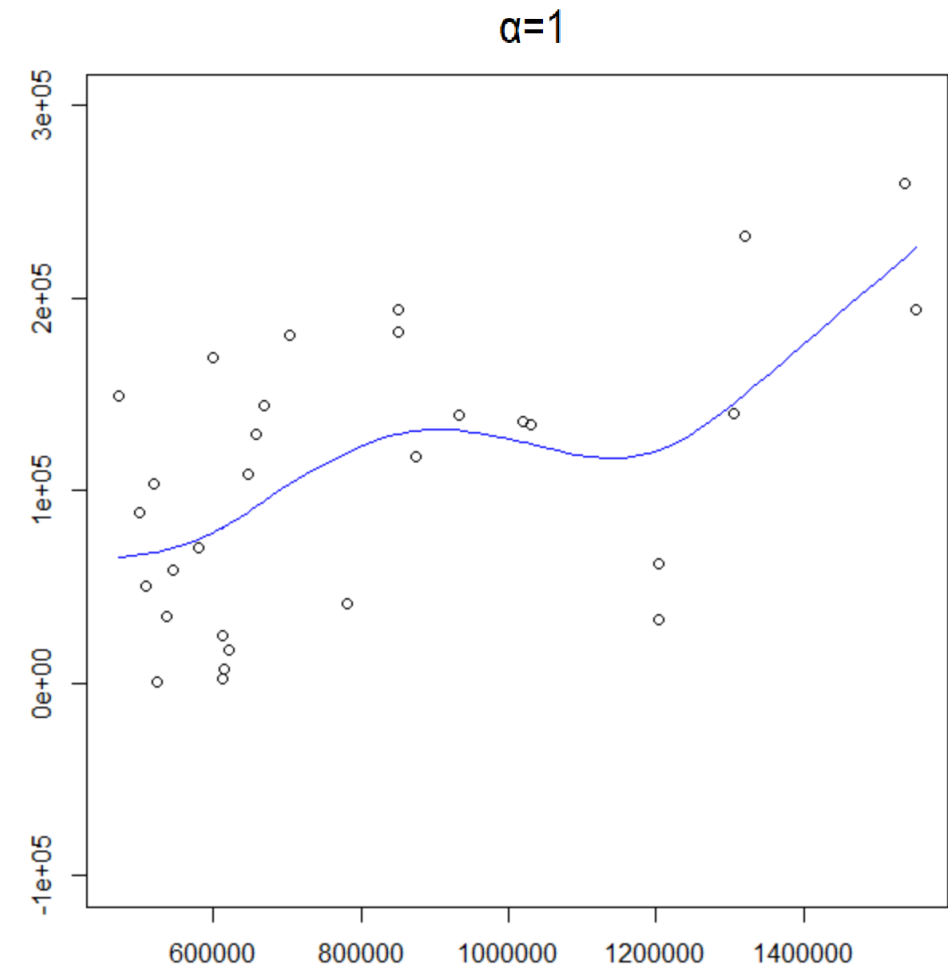
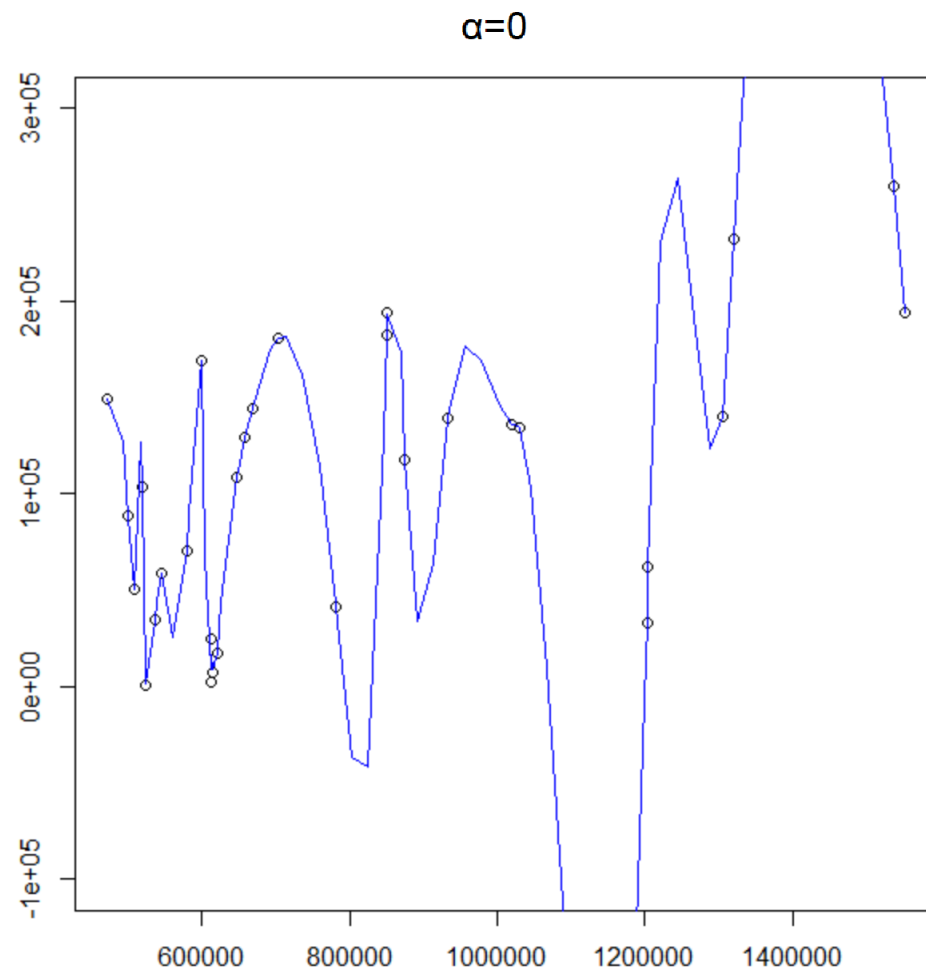
The “Least Squares with penalty” estimator is the solution of the optimization problem:

$$\inf_g SS(g)$$

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Obs: If α is small then regression curve will “follow” closely the statistical data.

If α is high, the regression curve will be smoother.



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Cubic splines

Splines provide a way to smoothly interpolate between fixed points.

Def. Suppose that $a < y_1 < y_2 < \dots < y_n < b$ and $g : [a, b] \rightarrow \mathbb{R}$.

The function g is called *cubic spline* if:

- g is a 3rd degree polynomial on each subinterval
 $(a, y_1), (y_1, y_2), \dots, (y_n, b)$
- g is continuous, with g' and g'' continuous on $[a, b]$.

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$\{y_i, i = 1, \dots, n\}$ are called knots.

Convention: $y_0 = a, y_{n+1} = b$.

A cubic spline has the form:

$$g(y) = d_i(y - y_i)^3 + c_i(y - y_i)^2 + b_i(y - y_i) + a_i, \quad \begin{array}{l} y_i \leq y \leq y_{i+1} \\ i = 0, \dots, n \end{array}$$

with the condition for continuity in $y_{i+1}, i = 0, \dots, n - 1$

$$d_i(y_{i+1} - y_i)^3 + c_i(y_{i+1} - y_i)^2 + b_i(y_{i+1} - y_i) + a_i = a_{i+1}.$$

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Def. A cubic spline over $[a,b]$ is called natural cubic spline (NCS) if its 2nd and 3rd order derivatives are null in a and b .

$$\left. \begin{array}{l} \text{Obs. From } g''(a) = g''(b) = 0 \\ g'''(a) = g'''(b) = 0 \end{array} \right\} \Rightarrow d_0 = c_0 = d_n = c_n = 0$$

$\Rightarrow g$ is linear on $[a, y_1]$ and $[y_n, b]$.

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Notations: $g_i = g(y_i)$, $i = 1, \dots, n$

$$\underline{g} = (g_1, \dots, g_n)'$$

$$\gamma_i = g''(y_i), \quad i = 1, \dots, n$$

($\gamma_1 = \gamma_n = 0$ because g is linear on $[a, y_1]$, $[y_n, b]$)

$$\underline{\gamma} = (\gamma_2, \dots, \gamma_{n-1})'$$

$$h_i = y_{i+1} - y_i, \quad i = 1, \dots, n-1$$

$$q_{ij} = \begin{cases} \frac{1}{h_{j-1}}, & i = j-1 \\ -\frac{1}{h_{j-1}} - \frac{1}{h_j}, & i = j \\ \frac{1}{h_j}, & i = j+1 \\ 0, & |i-j| \geq 2 \end{cases}$$

$$Q = \| q_{ij} \|_{\substack{i=1,\dots,n \\ j=2,\dots,n-1}}$$

$$\begin{cases} r_{ii} = \frac{1}{3}(h_{i-1} + h_i), & i = 2, \dots, n \\ r_{i,i+1} = r_{i+1,i} = \frac{1}{6}h_i, & i = 2, \dots, n-1 \\ r_{ij} = 0, & |i-j| \geq 2 \end{cases}$$

$$R = \| r_{ij} \|_{\substack{i=2,\dots,n-1 \\ j=2,\dots,n-1}}$$

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Proposition. *Function g can be specified by using $\underline{g}, \underline{\gamma}$ and $h_i, i = 1, \dots, n - 1$ as following:*

$$\left\{ \begin{array}{l} g(y) = g_1 - (y_1 - y) \left(\frac{g_2 - g_1}{h_1} - \frac{1}{6} h_1 \gamma_2 \right), \quad a \leq y \leq y_1, \\ \\ g(y) = \frac{1}{h_i} [(y - y_i) g_{i+1} + (y_{i+1} - y) g_i] - \\ - \frac{1}{6} (y - y_i)(y_{i+1} - y) \left[\left(1 + \frac{y - y_i}{h_i} \right) \gamma_{i+1} + \left(1 + \frac{y_{i+1} - y}{h_i} \right) \gamma_i \right], \\ \quad y_i \leq y \leq y_{i+1}, i = 1, \dots, n - 1 \\ \\ g(y) = g_n + (y - y_n) \left(\frac{g_n - g_{n-1}}{h_{n-1}} + \frac{1}{6} h_{n-1} \gamma_{n-1} \right), y_n \leq y \leq b. \end{array} \right.$$

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Theorem. Given the knots $a < y_1 < y_2 < \dots < y_n < b$ and $\underline{g}, \underline{\gamma}$ and $h_i, i = 1, \dots, n-1$, the function g built as described in the Proposition above is NCS iff $Q' \underline{g} = R \underline{\gamma}$.

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Corollary. If $Q' \underline{g} = R \underline{\gamma}$ then

$$\int_a^b (g''(y))^2 dy = \underline{\gamma}' R \underline{\gamma} = \underline{g}' K \underline{g}, \text{ where } K = QR^{-1}Q'$$

this is measure of the total curvature in $SS(g)$

Interpolation using NCS

$(x_i, y_i)_{i=1, \dots, n}$ the statistical data, $n \geq 2$

$a < y_1 < y_2 < \dots < y_n < b$ the knots

We search for a curve to fit $g(y_i) = x_i, i = 1, \dots, n$.

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Proposition. There is a unique NCS g , with the knots $y_1, \dots, y_n$ that satisfies $g(y_i) = x_i, i = 1, \dots, n$.

Algorithm for interpolation

Input: $(x_i, y_i)_{i=1, \dots, n}$

1. $g_i = x_i, i = 1, \dots, n$
2. with the previous notations, compute $x = Q' \underline{g}$
3. solve the system $R \underline{\gamma} = x \Rightarrow \underline{\gamma}$
4. having $\underline{g}, \underline{\gamma}$ and the knots, we built the NCS interpolant g according to the Proposition in slides 9-10.

Output: g

Nonparametric regression – curve fitting

Proposition. The NCS interpolant g is the solution of

$$\inf_{\tilde{g}} SS(\tilde{g})$$

$\tilde{g}$ interpolant

$$\tilde{g}(y_i) = x_i, i = 1, \dots, n$$

$\tilde{g}$ is differentiable

Obs. For any interpolant $\tilde{g}$, $SS(\tilde{g}) = \alpha \int_a^b (\tilde{g}''(y))^2 dy$
(because $\tilde{g}(y_i) = x_i, i = 1, \dots, n$).

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Smoothing using NCS

$(x_i, y_i)_{i=1, \dots, n}$ the statistical data, $n \geq 3$

$a < y_1 < y_2 < \dots < y_n < b$

The regression curve is

$$X = g(y) + \xi.$$

Proposition. If $\hat{g}$ is the solution of the optimization problem

$$\inf_{\substack{g \text{ is differentiable} \\ \text{on } [a, b]}} SS(g)$$

then $\hat{g}$ is NCS.

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Obs. Unlike the interpolation problem, here we don't know whether $\hat{g}(y_i) = x_i, i = 1, \dots, n$.

The Reinsch algorithm for smoothing using NCS

Input: $(x_i, y_i)_{i=1, \dots, n}, \alpha > 0$

1. Compute $z = Q' \cdot (x_1, \dots, x_n)'$
2. $R + \alpha Q'Q$ is symmetric and positive definite - it admits a Cholesky decomposition.

$R + \alpha Q'Q = LDL'$, where D is diagonal, positive definite and L has the properties:

$$\begin{cases} l_{ii} = 1 & \forall i = 1, \dots, n-2 \\ l_{ij} = 0 \text{ for } \forall j, 2 < j < i-2, i = 6, \dots, n-2 \end{cases}$$

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3. Solve the system $LDL'\underline{\gamma} = \underline{z}$ and we get $\underline{\gamma}$
4. Compute $\underline{\hat{g}} = (x_1, \dots, x_n)' - \alpha Q\underline{\gamma}$
5. Having $\underline{\hat{g}}, \underline{\gamma}$ and the knots $y_1, y_2, \dots, y_n$, we built the NCS smoothing function $\hat{g}$ as indicated in the Proposition in slides 9-10.

Output: $\hat{g}$ the smoothing function of the regression curve.

Obs. R and Q are the matrices defined in slide 8.

$R + \alpha Q'Q$ is decomposed into LDL' for computational purpose.